



AI-driven U.S. drought prediction using machine learning and deep learning

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Abstract

The importance of predicting drought, a significant environmental and socio-economic challenge, cannot be overstated. This research undertakes an extensive examination of drought phenomena, integrating two intricate datasets that detail weather and soil conditions across various counties in the United States (U.S.). We analyze data across the U.S. from 2000 to 2020, comparing algorithms that account for spatio-temporal structures. Specifically, we evaluate Gradient Boosting Machines techniques such as XGBoost, LightGBM, and CatBoost. Additionally, we develop deep learning (DL) models, including long short-term memory (LSTM), convolutional neural networks (CNN), and Transformer architectures. Hybrid models, such as CNN–LSTM and Attention–LSTM, are also explored. For ensemble methods, we implement stacking and voting classifiers to enhance robustness and performance. These models were chosen as they effectively capture spatio-temporal structures in the data, making them particularly suitable for drought prediction. We also perform clustering to divide the data into homogeneous regions and analyze feature importance using both Shapley Additive Explanations (SHAP) values and impurity-based methods derived from tree-based models (e.g., XGBoost, LightGBM, CatBoost), in order to identify the most predictive features of drought scores within each region. To the best of our knowledge and based on a comprehensive review of existing literature, our study is pioneering in utilizing these advanced methodologies for an in-depth analysis of drought score data in the U.S. Our findings indicate model accuracies ranging from 0.5938 to 0.9783, with the highest performance achieved by the Attention–LSTM Hybrid Model and the lowest by XGBoost, although this increased accuracy comes at the cost of significantly longer computation time for DL models.

Keywords Clustering · LSTM · CNN · Transformer · Hybrid and ensemble methods · Feature importance

1 Introduction

Drought is a prolonged period of significantly lower-than-average precipitation in a region, leading to water shortages. It can be categorized into different types based on its impact and underlying causes (Wilhite and Glantz 1985; Mishra and Singh 2010). Meteorological drought occurs due to a lack of precipitation over an extended period, relative to the long-term average for a specific region. Agricultural drought refers to insufficient soil moisture levels affecting crop growth and agricultural productivity, often

influenced by meteorological drought. Hydrological drought, on the other hand, is characterized by reduced water availability in surface and groundwater sources, such as rivers, lakes, and reservoirs, impacting water supply for ecosystems, industries, and human consumption. Unlike sudden natural disasters, drought develops over time and can persist for months or years, causing wide-ranging environmental and socio-economic impacts. The severity of a drought depends on its duration, intensity, and the region's vulnerability to water scarcity. Since the early 2000s, the western and southwestern regions of the United States (U.S.) have experienced multiple severe drought events, primarily due to below-normal precipitation (MacDonald 2010). These droughts have occurred intermittently over the years rather than as a single continuous drought period, with varying intensity and duration across different regions. The path of extratropical cyclones, influenced by climate phenomena such as the El Niño–Southern Oscillation, also plays a role

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in exacerbating drought conditions. The impact of global warming is expected to increase the frequency and severity of these droughts (IPCC 2021; Cai et al. 2014). Droughts can have more severe consequences than many other natural disasters due to their gradual onset and prolonged duration. California, for instance, faced an intense drought through 2010, which did not end until March 2011. This drought later expanded to the Southwest and Texas during the summer of 2011. In 2013 and early 2014, California experienced a resurgence of drought, impacting much of the western U.S. This period saw record low precipitation levels, with cities such as San Francisco receiving significantly less rainfall than usual. The winter snowpacks during 2012–2013 and 2013–2014 were among the lowest in the last century (Henry 2019). This drought also contributed to extensive wildfires in 2015, burning millions of acres across the western U.S. and Alaska, nearing national records. The prolonged drought has had a profound impact on water resources, agriculture, and natural ecosystems in these regions. Drought in the U.S., particularly in regions such as the Southwest, is closely linked to specific weather conditions and soil types. Weather patterns, such as El Niño, influence precipitation and temperature, affecting drought severity and duration (Williams et al. 2020). Soil types also play a crucial role; different soils retain or drain water differently, impacting how regions cope with low rainfall. For instance, sandy soils may lead to quicker moisture loss, exacerbating drought conditions, while clay soils might hold water longer but can become hard and impermeable when dry. This complex interaction between climate and soil impacts water availability, agriculture, and ecosystems (Sheffield et al. 2004).

The study of drought prediction and monitoring in the U.S. has evolved significantly, marked by a series of comprehensive research efforts. Luo and Wood (2007) enhanced the understanding of drought dynamics with advanced monitoring and predictive models, setting a benchmark for future studies. Similarly, Wu and Kinter (2009) explored the interplay between sea surface temperatures (SST), soil moisture, and drought occurrences, distinguishing the time scales of droughts and offering insights into the onset, duration, and severity of drought conditions based on SST variations and regional soil moisture levels. Various statistical and machine learning (ML) methods have been employed in drought prediction. Early studies primarily relied on statistical regression models and time series analysis, using indices such as the Standardized Precipitation Index (SPI) and Standardized Precipitation Evapotranspiration Index (SPEI) to quantify drought severity across different timescales (McKee et al. 1993; Vicente-Serrano et al. 2010). These models provided valuable insights but struggled with capturing nonlinear dependencies and complex spatial-temporal relationships in climate data. Later studies introduced probabilistic modeling approaches, such as the Standardized Soil Moisture

Index (SMI) framework, which accounted for soil moisture variability and provided more physically meaningful drought characterizations (AghaKouchak 2014; Saharwardi et al. 2025). While probabilistic models improved uncertainty quantification, they often required extensive historical data and assumptions about climate stationarity, limiting their applicability under changing climate conditions. With advances in computational methods, ML techniques have gained prominence in drought forecasting. Felsche and Ludwig (2021) demonstrated the effectiveness of ML models, such as Random Forest (RF) and Gradient Boosting Machines (GBM), for pattern recognition in climate datasets. These approaches successfully identified complex climatic interactions but required large amounts of labeled training data and suffered from interpretability issues. More recent studies have explored deep learning (DL) architectures, particularly convolutional neural networks (CNNs) and long short-term memory (LSTM) networks, which can capture spatial and temporal dependencies in climate variables. Brust et al. (2021) developed “DroughtCast,” a deep learning-based framework that leveraged LSTM networks to enhance long-term drought forecasting accuracy. While DL models outperformed traditional methods, they remained computationally expensive and required substantial tuning to generalize across diverse climate conditions. Despite these advancements, existing drought prediction models exhibit several limitations. Many ML approaches lack explainability, making it difficult to interpret the underlying drivers of drought. Additionally, data sparsity in historical climate records affects model reliability, particularly in arid regions. Furthermore, most models focus on short-term prediction horizons, with limited ability to generalize under future climate scenarios. Recent studies emphasize the need for hybrid approaches that combine physical climate models with data-driven techniques to improve both accuracy and interpretability (Saharwardi et al. 2025).

Our study builds on these foundations by integrating two intricate datasets detailing weather and soil conditions across various U.S. counties from 2000 to 2020. We compare spatio-temporal algorithms, including Gradient Boosting Machine (GBM) techniques such as XGBoost, LightGBM, and CatBoost, which build decision trees sequentially—each tree correcting the errors of the previous one—thereby enhancing prediction accuracy (Brust et al. 2021). Additionally, we develop DL models, including LSTM, a type of recurrent neural network well-suited for time series data capable of learning long-term dependencies and patterns over time; CNN, typically used for spatial data to extract important features from input data; and Transformers, which use self-attention mechanisms to weigh the importance of different parts of the input data, making them effective for capturing both short and long-term dependencies in time series data (Li et al. 2021; Yu et al. 2019; Han et al. 2021).

We also explore hybrid models, such as CNN–LSTM, combining CNN layers to extract spatial features and LSTM layers to capture temporal dependencies, and Attention-LSTM, incorporating attention mechanisms into LSTM networks to improve interpretability and performance of the predictions (Vo et al. 2023). For ensemble methods, we implement stacking, combining multiple models (XGBoost, LightGBM, and CatBoost) and using another model to learn how to best combine their predictions, and a voting classifier, aggregating predictions from multiple models to enhance robustness and performance (Kumari et al. 2021). These models were specifically selected for their ability to capture both spatial and temporal dependencies inherent in drought prediction, as they effectively process lagged features and differences across time steps. Their ability to learn intricate spatio-temporal relationships enables better representation of climate patterns and soil dynamics, which are crucial for improving predictive accuracy (Vo et al. 2023).

Unique to our approach, we perform clustering to divide the data into homogeneous regions with respect to variation in features and analyze feature importance within each region to identify the most predictive features of drought scores. This regional analysis allows us to tailor drought prediction models more accurately to specific areas.

2 Data overview and preprocessing

Two distinct datasets have been compiled, encompassing comprehensive details on both weather and soil conditions. The weather dataset encompasses critical parameters such as temperature, wind speed, humidity, surface pressure, frost point, and precipitation between 2000 to 2020. This dataset is organized with a temporal dimension, specifying values for specific dates across various counties in the U.S. The weather features are recorded at a daily resolution, allowing for high-frequency analysis of weather patterns. Additionally, drought scores are included and observed weekly, providing a comprehensive view of weather conditions and their impact over time. Since drought scores are observed on a weekly basis, we aggregated the daily climate data into weekly averages to ensure temporal alignment between the datasets. This approach maintains consistency in the analysis and enables a more accurate assessment of the relationship between weather patterns and drought conditions over time.

Besides, the soil dataset includes a diverse array of information, encompassing nutrient levels, rooting characteristics, oxygen content, salinity, toxicity levels, workability, slope, aspect, water characteristics, and land properties pertaining to cultivation and forested areas. These soil attributes are meticulously documented for each county, with accompanying geographical coordinates in the form of longitude and latitude. The soil data is mostly static, as it

represents long-term soil properties that do not change significantly over short periods. However, certain parameters, such as mapped water bodies, barren or sparsely vegetated land, grass/scrub/woodland cover, irrigated cultivated land, nutrient availability, can exhibit seasonal or interannual variability. These changes are influenced by factors such as precipitation patterns, irrigation, agricultural practices, and land management decisions. Despite these fluctuations, soil data is not updated as frequently as weather data. The inclusion of such detailed soil information provides a comprehensive understanding of the diverse soil compositions and characteristics present across different regions.

Meteorological drought is determined by the lack of precipitation and how conditions such as temperature and winds affect the amount of moisture. It is expressed in relation to the average conditions for a region and is region-specific since precipitation is highly variable from region to region. Agricultural drought links the characteristics of meteorological drought to agriculture or landscapes, focusing on precipitation shortages, evaporative demand, and soil moisture deficits (Lal 2019). This type of drought is also dependent upon plant type, stage of growth, and soil properties. In contrast, hydrological drought refers to the prolonged reduction in water availability in surface and groundwater sources, such as rivers, reservoirs, and aquifers (Van Lanen 2006). Unlike meteorological and agricultural droughts, which primarily affect short-term precipitation patterns and soil moisture, hydrological drought develops over a longer time-scale and has significant consequences for water resource management, irrigation, and ecosystem sustainability. It can persist even after meteorological conditions improve, as water bodies take longer to recover from extended dry periods (Mishra and Singh 2010). Hydrological drought is particularly critical for regions that rely heavily on groundwater and surface water for agriculture and drinking water supply, as extended depletion of these resources can lead to severe socio-economic impacts (Huang et al. 2017; Hisdal et al. 2001).

We will take meteorological, agricultural, and hydrological drought into account, as the datasets contain sufficient information related to these drought types. While meteorological and agricultural droughts can be assessed through precipitation, temperature, and soil moisture variables, hydrological drought is analyzed using long-term water availability data. Data on the Drought Severity and Coverage Index (DSCI) is also provided and it is classified into six categories.¹

¹ For more information about drought classification, visit: <https://droughtmonitor.unl.edu/About/AbouttheData/DroughtClassification.aspx>

Descriptions of the datasets are provided in Table 1. The integration of these two datasets is facilitated through the utilization of county Federal Information Processing Standards (FIPS) codes, with identical codes present in both the weather and soil datasets. The dataset is structured at the county level, with 5-digit FIPS codes, where the first two digits represent the state and the last three represent the county. It includes latitude, longitude, and meteorological variables, requiring aggregation to the state level when analyzing broader regional trends. The datasets collectively encompass information from 49 states. The merged dataset, spanning the years from 2000 to 2020, comprises a substantial 3,406,368 observations. This dataset is characterized by 53 features, each contributing unique information. The

extensive size and comprehensive range of features make this dataset a valuable resource for in-depth analysis and exploration, particularly in the context of predicting drought scores based on the provided weather and soil features.

The primary objective is the prediction of drought scores, categorized on a 0 to 5 scale, which follows the classification system of the U.S. Drought Monitor (USDM) summarized in Table 2. A score of 0 represents no drought, indicating normal or wet conditions, corresponding to SPI and SPEI of -0.49 or above. A score of 1 corresponds to D0 (Abnormally Dry), indicating short-term dryness that may slow crop and pasture growth, mapped to an SPI and SPEI range of -0.5 to -0.79 . A score of 2 aligns with D1 (Moderate Drought),

Table 1 Climate and soil features

Climate features	Soil features
U.S. county FIPS code	U.S. county FIPS code
Observation date (weekly)	Latitude and Longitude
Precipitation (mm day^{-1})	Median elevation (meters)
Surface pressure (kPa)	Slope
Specific humidity at 2 ms (g/kg)	North: $0^\circ < \text{aspect} \leq 45^\circ$ or $315^\circ < \text{aspect} \leq 360^\circ$
Temperature at 2 ms ($^\circ\text{C}$)	East: $45^\circ < \text{aspect} \leq 135^\circ$
Dew/Frost point at 2 ms ($^\circ\text{C}$)	South: $135^\circ < \text{aspect} \leq 225^\circ$
Wet bulb temperature at 2 ms ($^\circ\text{C}$)	West: $225^\circ < \text{aspect} \leq 315^\circ$
Maximum temperature at 2 ms ($^\circ\text{C}$)	Mapped water bodies
Minimum temperature at 2 ms ($^\circ\text{C}$)	Barren/very sparsely vegetated land
Temperature range at 2 ms ($^\circ\text{C}$)	Built-up land (residential and infrastructure)
Earth skin temperature ($^\circ\text{C}$)	Grass/scrub/woodland
Wind speed at 10 ms (m/s)	Forest land
Maximum wind speed at 10 ms (m/s)	Irrigated cultivated land, according to GMIA 4.0
Minimum wind speed at 10 ms (m/s)	Total cultivated land
Wind speed range at 10 ms (m/s)	Nutrient availability and retention capacity
Wind speed at 50 ms (m/s)	Rooting conditions
Maximum wind speed at 50 ms (m/s)	Oxygen availability to roots
Minimum wind speed at 50 ms (m/s)	Excess salts
Wind speed range at 50 ms (m/s)	Toxicity
Measure of drought ranging from 0 to 5	Workability (constraining field management)

Table 2 Drought classification based on SPI and SPEI

	Category	Description	Values for SPI and SPEI
0	None	Normal or wet conditions	-0.49 or above
1	D0	Abnormally dry	-0.5 to -0.79
2	D1	Moderate drought	-0.8 to -1.29
3	D2	Severe drought	-1.3 to -1.59
4	D3	Extreme drought	-1.6 to -1.99
5	D4	Exceptional drought	-2.0 or less

This table is taken from USDM, <https://droughtmonitor.unl.edu/About/AbouttheData/DroughtClassification.aspx>

signifying more prolonged dryness with potential crop damage and water shortages, occurring within an SPI and SPEI range of -0.8 to -1.29 . A score of 3 represents D2 (Severe Drought), leading to significant crop losses and widespread water shortages, with an SPI and SPEI between -1.3 and -1.59 . A score of 4 corresponds to D3 (Extreme Drought), indicating major crop and pasture losses alongside widespread water restrictions, mapped to an SPI and SPEI range of -1.6 to -1.99 . Finally, a score of 5 corresponds to D4 (Exceptional Drought), the most severe condition, characterized by extensive crop failures and severe water shortages, often resulting in emergency conditions, with an SPI and SPEI of -2.0 or lower.

This classification system directly follows the USDM thresholds, ensuring consistency with established drought severity definitions used in climate monitoring and policy applications. The dataset distribution reflects the prevalence of different drought levels, with no drought (score 0) occurring in approximately 49.12% of cases, while D0 (score 1) and D1 (score 2) account for 21.05% and 14.04%, respectively. The more extreme drought conditions-D2 (Severe), D3 (Extreme), and D4 (Exceptional)-are observed less frequently, making up 10.53%, 3.51%, and 1.75% of cases, respectively, underscoring the importance of proactive drought management strategies to mitigate their impact. (Please refer to Appendix 1 for visualization of percent drought scores.) The predictive modeling process will leverage the diverse features provided in the integrated datasets, spanning climatic factors from the weather data and soil characteristics from the soil dataset. The outcome of this analysis aims to discern patterns and relationships between these features and the corresponding drought scores, contributing to a more nuanced understanding of drought conditions across the specified geographic regions.

Figure 1 shows the percentage of the U.S. affected by various levels of drought from 2000 through 2020. Different colors represent the intensity of drought, ranging from (abnormally dry) to (exceptional drought). Over time, the peaks and valleys suggest fluctuating periods of drought severity, with certain years experiencing significantly higher coverage of extreme to exceptional drought conditions such as 2000, 2008 and 2012.

In Fig. 2, a geographical representation illustrates the drought scores for each county in the U.S. counties are color-coded, with shades of light beige indicating areas without drought conditions and varying intensities of dark red to black representing higher levels of drought. Upon closer examination of the map, it becomes apparent that certain states, including Nevada, Idaho, Arizona, California, Georgia, Kansas, Montana, Texas, Colorado, Wyoming, Oklahoma, and New Mexico, are particularly susceptible to drought risk. This visual depiction effectively identifies and highlights regions with varying degrees of vulnerability to drought across these states. This observation aligns with existing literature on drought-prone regions in the U.S. The identified states consistently emerge as areas of heightened concern for drought conditions, reinforcing the significance of targeted interventions and adaptive strategies in these regions.

An analysis of weather conditions across the U.S. reveals distinct regional patterns (Fig. 3). In the southern coastal regions, particularly in Florida and Texas, higher temperatures are observed due to their proximity to the equator. Conversely, in the western and southwestern regions, notably California, Nevada, Arizona, and parts of Oregon and Washington, significantly lower humidity and precipitation levels are recorded, leading to arid conditions. However, it is crucial to distinguish water-stressed

Fig. 1 Percentage of the U.S. drought for each year

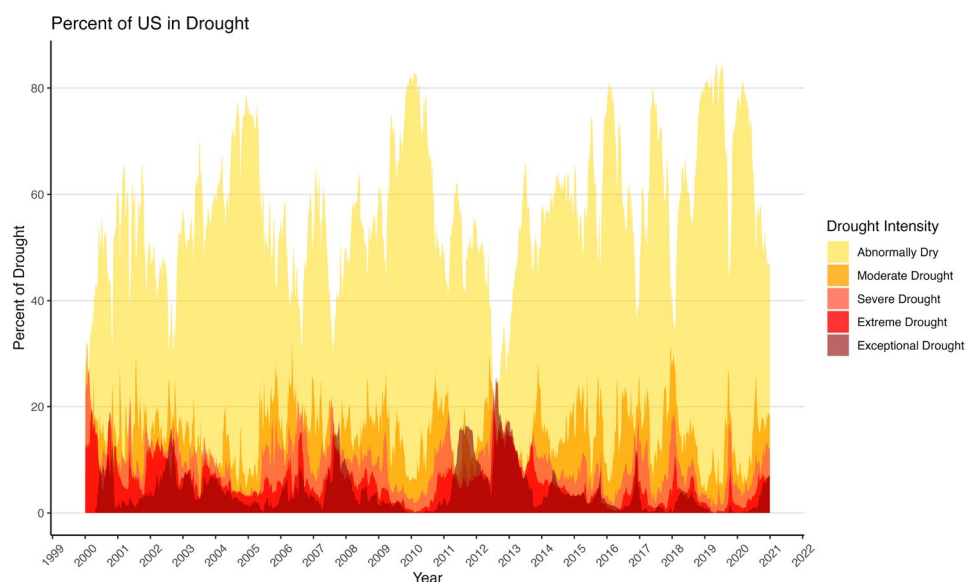


Fig. 2 U.S. map colored by drought score

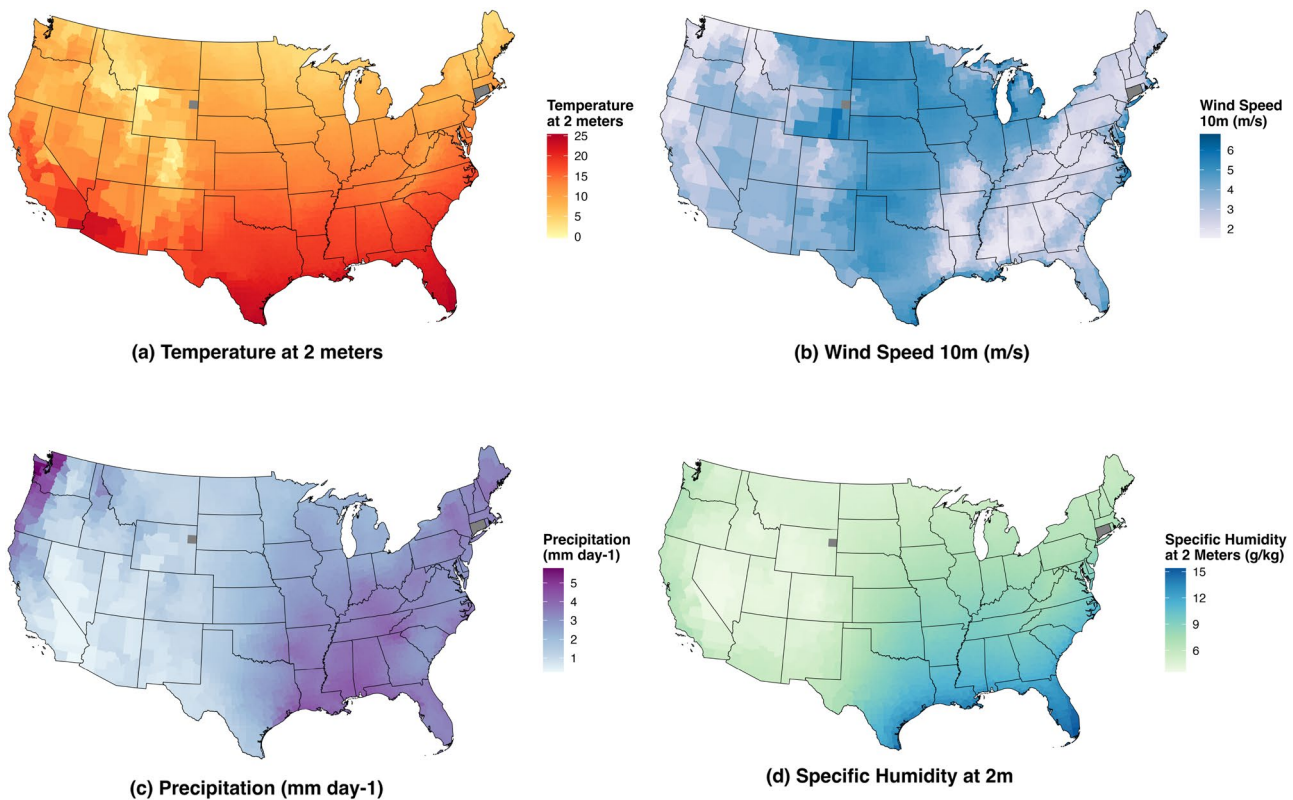
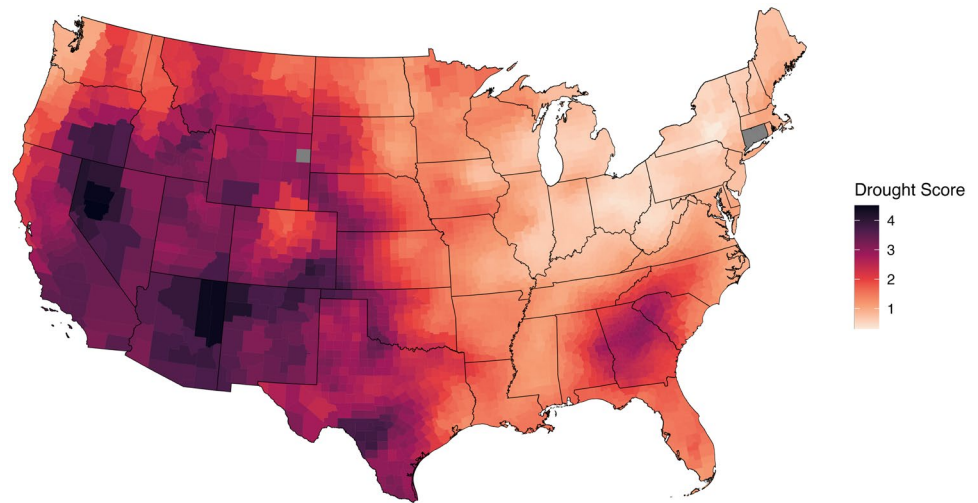


Fig. 3 U.S. map colored by weather conditions; **a** temperature, **b** wind speed, **c** precipitation, **d** humidity

regions from actual drought anomalies, as aridity alone does not necessarily indicate ongoing drought. A strong inverse relationship is observed between humidity, precipitation, and drought severity, particularly in the western and central regions, where decreasing humidity and precipitation correlate with higher drought scores. This finding is statistically validated by a Granger causality analysis, which shows that past values of precipitation ($p < 10^{-141}$)

and specific humidity ($p < 10^{-63}$) significantly predict future drought conditions.

Furthermore, wind speed exhibits notable spatial trends, with higher wind speeds predominantly observed in the central U.S. and parts of the Northwest, while lower wind speeds are observed in the Southeast and Northeast. A one-way Analysis of Variance (ANOVA) test confirms

that wind speeds in these regions are significantly higher than in other parts of the country ($p < 0.05$), validating the observed spatial differences. This suggests that atmospheric dynamics in these regions contribute to increased wind activity, which may exacerbate drought conditions. Additionally, a Granger causality test reveals a significant relationship between maximum wind speed at 50 m ($p < 10^{-300}$), maximum wind speed at 10 m ($p < 10^{-295}$), and drought severity. This suggests that stronger winds may contribute to drought intensification by accelerating evapotranspiration, reducing soil moisture, and increasing surface drying rates.

Temperature also plays a significant role in drought formation. The Granger causality analysis indicates that air temperature at 2 m ($p < 10^{-300}$), maximum air temperature at 2 m ($p < 10^{-300}$), and minimum air temperature at 2 m ($p < 10^{-300}$) are strong predictors of drought scores. Higher temperatures contribute to increased evaporation, exacerbating soil moisture loss and intensifying drought conditions over time. These findings highlight the crucial role of thermal factors in shaping drought dynamics.

In Fig. 4, specific soil conditions are depicted, providing insights into the distribution of certain land features. The data indicates a generally lower prevalence of cultivated land in the western regions, with some exceptions such as parts of California. Forested areas are notably abundant in the eastern coastal regions and significantly present in the northwestern coastal areas, particularly in Washington and Oregon. It is observed that cultivated land is inversely related to drought conditions, with lower levels of cultivation corresponding to reduced drought severity. Similarly, in forested areas, lower levels of drought are noted. These findings suggest a geographical correlation between land use patterns and drought conditions. The lower prevalence of cultivated land in the western U.S., compared to the Midwest and Great Plains, may contribute to variations in drought susceptibility, while the abundance of forested areas in the east, particularly along the coast, may also play a role in mitigating drought

effects. This nuanced understanding of soil conditions and land use contributes valuable insights to the broader context of regional vulnerability to drought, providing a basis for further investigation and potential interventions.

Prior to modeling, all numeric features are scaled for the pattern discovery and analysis part. Robust scaling is specifically employed using the equation (1) for certain variables due to the presence of a substantial number of outliers. This choice is made because robust scaling relies on the median (M) and the interquartile range (IQR), which are known for their resilience against the influence of outliers. The presence of outliers can be observed in Appendix 2, where the boxplot visualization highlights extreme values across multiple features.

$$\left(\frac{X - \text{Median}(X)}{\text{IQR}(X)} \right). \quad (1)$$

2.1 Cross validation

Cross-validation is a method used for evaluating and comparing learning algorithms by dividing data into three segments: one used to learn or train a model, second group used for predicting model and the other used to validate the model. In the process of model selection, hyperparameters are fine-tuned using a validation set to determine the combination that yields the best performance, typically measured by metrics such as accuracy. The training set is used to train the model, while the validation set acts as a proxy for unseen data to evaluate the model's performance. Various hyperparameter values are systematically tested, and the model's performance on the validation set is assessed. The set of hyperparameters that results in the highest performance metric on the validation set is selected as the optimal configuration. Table 3 provides a breakdown of dataset sizes allocated to training, test, and validation sets.

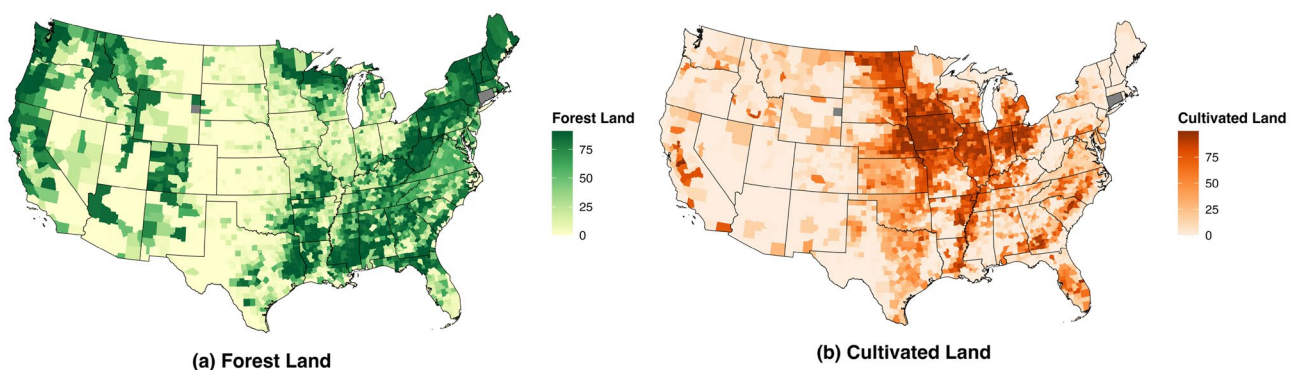


Fig. 4 U.S. map colored by soil conditions; **a** forest land, **b** cultivated land

Table 3 Distribution of the dataset into training, test and validation sets

	Number of observations	Starting date	Ending date
Training data	2,756,796	2000-01-04	2016-12-27
Validation data	323,232	2017-01-01	2018-12-31
Test data	326,340	2019-01-01	2020-12-31

2.2 Clustering

When assessing drought in the U.S., it is essential to consider the varying characteristics and drought patterns across different regions. Each region has distinct climate and soil properties, which can significantly influence the performance of predictive models (Batmaz 2022). Therefore, an effective approach is to first divide the data into clusters (representing different regions) and then perform separate and more precise modeling for each cluster. This strategy allows for the identification of the most suitable models for each region and the determination of which features are most significant for those models. By tailoring the models to the specific conditions of each region, we can enhance the accuracy and relevance of drought predictions and gain deeper insights into regional drought dynamics (Na et al. 2010).

To determine the optimal number of clusters, we used both the Elbow Method and the Silhouette Method shown in Fig. 5. The Elbow Method involves plotting the total within-cluster sum of squares (WSS) against the number of clusters (k) (Syakur et al. 2018). The plot shows a distinct elbow at $k = 3$, where the decrease in WSS slows down, indicating that adding more clusters beyond this point provides diminishing returns. The Silhouette Method measures how similar each point is to its own cluster compared to other clusters (Saputra et al. 2020), with the average silhouette width plotted against k . The highest silhouette score

at $k = 3$ suggests that three clusters offer well-defined and well-separated groups. Both methods consistently indicate that three clusters are optimal, balancing model simplicity and explanatory power.

K-means clustering is applied to the dataset to group the data points into three clusters. This unsupervised learning method assigns each data point to one of the three clusters based on similarity, minimizing the within-cluster variance. Principal Component Analysis (PCA) is used to reduce the dimensionality of the dataset to two dimensions for visualization purposes. PCA helps to project the high-dimensional data into a 2D plane while retaining most of the variance. In Fig. 6, three clusters can be interpreted as three distinct regions in the U.S. based on the dataset's features (such as weather and soil conditions). Each region (cluster) groups together counties with similar characteristics.

The map of the U.S. in Fig. 7 is divided into three clusters, represented by different colors, based on climate and soil characteristics using k-means clustering. Cluster 1 (red) is predominantly spread across the western and northern parts of the U.S., including large portions of the Midwest,

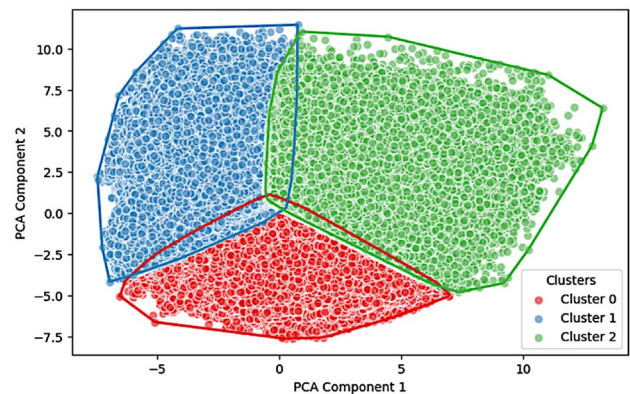
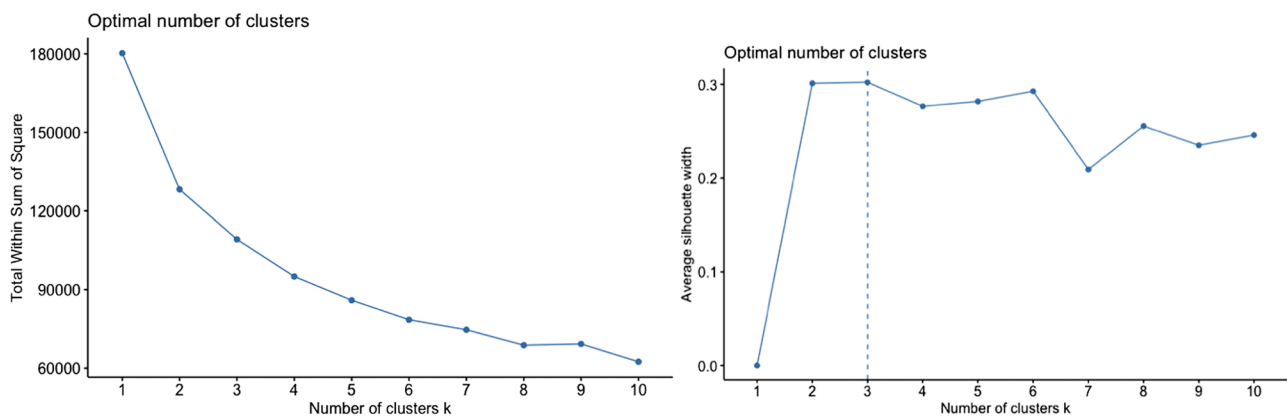
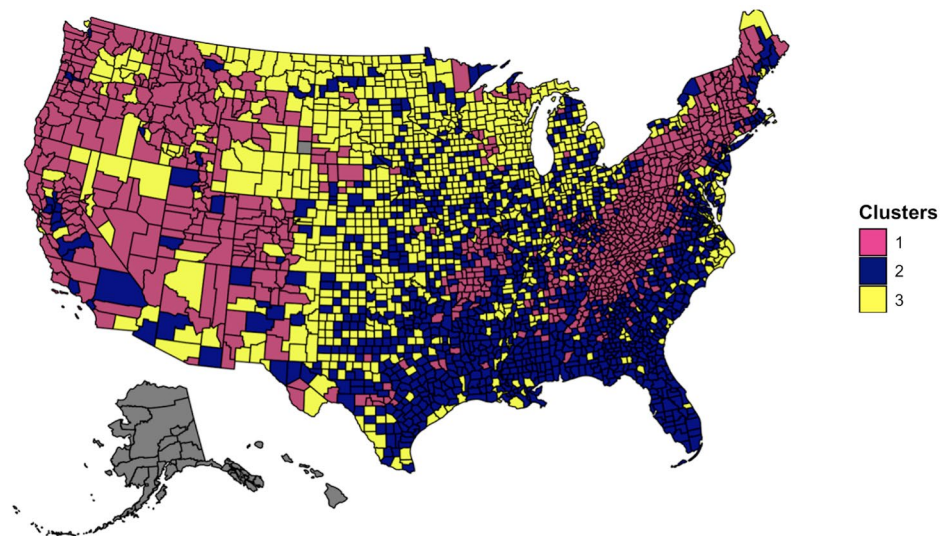
**Fig. 6** Cluster plot**Fig. 5** Optimal number of clusters according to WSS and Silhouette width

Fig. 7 U.S. map colored by clusters



rather than being concentrated in the Northeast as initially described. The West Coast also contains significant areas from Cluster 1, though it is mixed with other clusters in some regions. Cluster 2 (blue) spans much of the Southeast and parts of the Midwest, accurately reflecting regions with warmer temperatures and higher humidity. This classification correctly aligns with the southeastern U.S., covering states like Georgia, Alabama, and South Carolina. Cluster 3 (yellow) is widely distributed across the central U.S. and parts of the Southwest, particularly in the Great Plains and western regions, indicating drier conditions and temperature variations. This clustering allows for targeted and region-specific analysis of drought patterns and predictions, enabling the development of customized strategies for drought monitoring and management tailored to the unique characteristics of each region.

By using these clusters, models are evaluated within each region, applying and examining the models separately for each area to gain deeper insights and more accurate predictions specific to each cluster.

3 Implementation of the algorithms

We are addressing a classification problem where the drought score is the categorical target variable with levels ranging from 0 to 5. To tackle this, a variety of ML, DL, and hybrid models are implemented. The ML models include techniques such as XGBoost, LightGBM, CatBoost, which sequentially build trees to correct errors from previous ones. The DL models encompass architectures such as LSTM, CNN, and Transformers. Drought prediction is a complex task that requires capturing both spatial and temporal dependencies in the data. To address this, we selected models that excel

in different aspects of learning from spatio-temporal data. GBMs are highly effective at capturing non-linear relationships between climate variables and drought scores while being computationally efficient for structured tabular data (Chen and Guestrin 2016; Ke et al. 2017; Dorogush et al. 2018; Brust et al. 2021). LSTM is particularly well-suited for handling long-short term dependencies in time series, allowing it to learn complex temporal patterns critical for drought forecasting (Hochreiter and Schmidhuber 1997; Yu et al. 2019; Vo et al. 2023). Transformer models, on the other hand, leverage self-attention mechanisms to weigh the importance of different parts of the input, making them powerful in capturing both spatial and temporal relationships simultaneously (Vaswani et al. 2017; Han et al. 2021; Amanambu et al. 2022). Their ability to process long-range dependencies efficiently gives them an advantage over traditional recurrent networks, making them particularly valuable in modeling large-scale climate patterns. These choices ensure that our approach is tailored to the unique spatio-temporal nature of drought prediction. Additionally, hybrid models are explored, such as CNN-LSTM, which combines CNN layers for extracting spatial features with LSTM layers for capturing temporal dependencies, and Attention-LSTM, which incorporates attention mechanisms into LSTM networks to improve focus on important time steps (Krasnopolsky and Fox-Rabinovitz 2006). Ensemble methods such as Stacking and Voting Classifier are also implemented. Stacking combines multiple models and uses another model to learn how to best combine their predictions, while Voting Classifier aggregates predictions from multiple models to enhance robustness and performance (Kumari et al. 2021; Pavlyshenko 2018).

For evaluating the performance of these models, metrics such as accuracy, sensitivity (recall), specificity, and F1

score are used. Accuracy measures the overall correctness of the model, sensitivity assesses the model's ability to correctly identify positive instances, specificity evaluates the model's ability to correctly identify negative instances, and the F1 score provides a balance between precision and recall, especially useful in cases of class imbalance.

By implementing and evaluating these models across different clusters, we aim to determine the most effective models for each region and examine which climate factors and soil properties are affected on drought severity.

The experiments in this study are conducted on a MacBook Pro (Apple M3 Pro), which features a 12-core CPU, comprising 6 performance cores and 6 efficiency cores, optimized for handling both high-performance and energy-efficient tasks. The system is equipped with 18 GB of RAM, ensuring sufficient memory for computationally demanding ML and DL models. The study is performed on macOS Sonoma 14.x, utilizing Python (Version 3.8) as the primary programming language. Several essential libraries are employed to implement and evaluate the models, including scikit-learn for traditional ML algorithms, statsmodels for statistical modeling, and TensorFlow for DL frameworks.

3.1 XGBoost

The XGBoost works by constructing an ensemble of decision trees sequentially, where each tree attempts to correct the errors of the previous ones (Chen et al. 2015). The XGBoost model for drought prediction is designed to classify drought scores (ranging from 0 to 5) across different regions in the U.S. The data is first divided into three clusters based on climate properties using k-means clustering, ensuring region-specific modeling. A grid search is then performed separately for each cluster to identify the optimal hyperparameters, including learning rate, max depth, subsample, `colsample_bytree`, and the number of estimators. The learning rate adjusts the speed of learning by modulating the weight of new trees, with lower rates helping to avoid overfitting. The `n_estimators` parameter defines the number of boosting rounds, essentially the number of trees in the model. Max depth controls the depth of each tree, balancing model complexity and the ability to capture intricate data patterns. Subsample and `colsample_bytree` reduce model variance by limiting the fraction of the dataset and features used for each tree. The gamma parameter acts as a regularizer, requiring a minimum loss reduction for node splitting to prevent overfitting. Regularization terms `reg_alpha` and `reg_lambda` apply L1 and L2 penalties to the model weights, further curbing complexity. `scale_pos_weight` addresses class imbalance by balancing the positive and negative class weights. These hyperparameters are fine-tuned using grid search which

optimizes the model's performance. The optimal hyperparameters for XGBoost, identified through grid search on the validation set, are summarized in Table 4. This tuning process ensures the model is well-suited for accurate and robust drought prediction across different regions.

3.2 CatBoost

CatBoost is a gradient boosting framework designed specifically for handling categorical features efficiently and effectively. Unlike other gradient boosting implementations, CatBoost automatically handles categorical variables, reducing the need for extensive preprocessing and feature engineering. It uses a technique called ordered boosting, which aims to reduce overfitting by incorporating random permutations of the dataset (Hancock and Khoshgoftaar 2020). CatBoost also employs efficient and scalable algorithms to ensure fast training and prediction times, making it suitable for large datasets. Its robust handling of categorical data, combined with powerful gradient boosting methods, often results in superior performance on datasets with a mix of numerical and categorical features.

The CatBoost model hyperparameters are selected through an exhaustive grid search process conducted individually for each cluster. Despite the distinct characteristics of each region, the optimal values for all clusters converge to the same set of hyperparameters. The grid search explores variations in `iterations`, `depth`, `learning_rate`, and `l2_leaf_reg` to balance model complexity and predictive accuracy. The best-performing configuration includes `iterations` set to 500, ensuring sufficient boosting rounds for learning, and `depth` set to 10, allowing the model to capture intricate patterns without excessive overfitting. A `learning_rate` of 0.01 provides stable convergence, while `l2_leaf_reg` mitigates overfitting by applying L2 regularization. The fact that the same hyperparameters yield optimal performance across all clusters suggests that CatBoost's learning process generalizes well to different regional drought patterns while maintaining high predictive accuracy.

Table 4 Selected hyperparameters for XGBoost

Hyperparameters	Cluster 1	Cluster 2	Cluster 3
<code>learning_rate</code>	0.2	0.1	0.2
<code>n_estimators</code>	250	100	300
<code>max_depth</code>	7	20	10
<code>subsample</code>	0.6	0.2	0.3
<code>gamma</code>	3	3	3
<code>reg_alpha</code>	0.2	0.01	0.1
<code>reg_lambda</code>	20	12	20

3.3 Light gradient boosting machine (LightGBM)

LightGBM is a highly efficient gradient boosting framework designed for speed and performance, particularly with large datasets and high-dimensional data. Unlike traditional boosting algorithms, LightGBM uses a leaf-wise growth strategy, which tends to reduce loss more effectively by expanding the most promising leaf nodes. This approach, combined with a histogram-based algorithm for feature binning, significantly accelerates training and reduces memory usage (Ke et al. 2017). LightGBM also natively supports categorical features, enhancing its utility in diverse datasets. These characteristics make LightGBM an ideal choice for ML tasks requiring high accuracy and computational efficiency.

Key hyperparameters of LightGBM include `boosting_type`, `max_bin`, `learning_rate`, `n_estimators`, and `num_leaves` as summarized in Table 5. The `boosting_type` parameter specifies the boosting method, 'dart' (Dropouts meet Multiple Additive Regression Trees) is used for better generalization and reduced overfitting. The `max_bin` parameter controls the maximum number of bins for feature discretization, with a larger value allowing for finer discretization at the cost of increased computation. The `learning_rate` is set low (0.01) to ensure gradual learning, while the `n_estimators` is set high (1000) to allow the model sufficient iterations to converge. The `num_leaves` parameter, set to a large value (128), determines the maximum number of leaves in one tree and helps control the model's complexity.

Table 5 Selected hyperparameters for LightGBM

Hyperparameters	Cluster 1	Cluster 2	Cluster 3
<code>boosting_type</code>	Dart	Dart	Dart
<code>max_bin</code>	250	255	250
<code>learning_rate</code>	0.01	0.01	0.01
<code>n_estimators</code>	1000	1000	1000
<code>num_leaves</code>	120	128	120

3.4 Ensemble methods

Ensemble methods combine multiple ML models to produce a single optimal predictive model. The goal is to leverage the strengths of each individual model to improve overall performance. Ensemble methods can help reduce overfitting, increase accuracy, and provide more robust predictions. In the context of drought prediction using climate data, ensemble methods can integrate different models that capture various aspects of climate patterns, leading to more accurate and reliable predictions. Prior to applying ensemble techniques, individual machine learning models were built separately for each cluster, with hyperparameters carefully tuned to optimize regional performance. These same tuned hyperparameters were then utilized within the ensemble models to ensure consistency and preserve cluster-specific optimization.

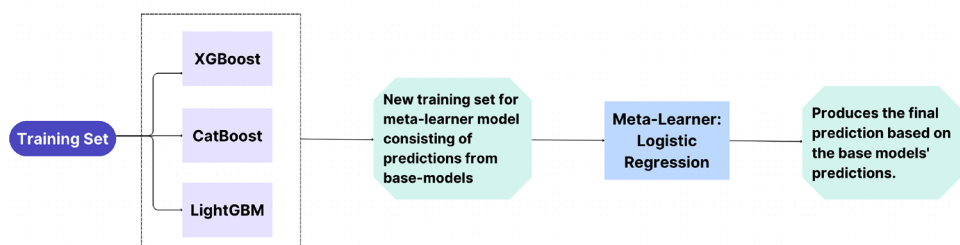
3.4.1 Stacking

Stacking is an ensemble technique that involves training multiple base models and then using their predictions as input features for a higher-level meta-learner model (Pavlyshenko 2018). In the context of drought prediction, base models such as LightGBM, CatBoost, and XGBoost are trained on climate and soil types datasets, such as precipitation, temperature, humidity, and wind speed. Once these models are trained, a meta-learner (simpler model), logistic regression, is trained on the predictions made by the base models. This process aims to learn how to best combine the predictions of the base models to improve overall performance (Ghasemian et al. 2020) (Fig. 8).

3.4.2 Voting classifier

A voting classifier is another ensemble method where multiple models are trained separately, and their predictions are combined to make a final decision (Kumari et al. 2021). In the context of drought prediction, models such as LightGBM, CatBoost, and XGBoost are used as base classifiers. We apply soft voting, where the predicted probabilities of each base model are averaged, and the class with the highest average probability is selected as the final prediction.

Fig. 8 Stacking model architecture



Soft voting leverages the strengths of multiple models by combining their probability predictions, which often leads to more accurate and stable results compared to individual models. This approach is particularly useful in drought prediction, where combining different models can help to account for the complex and multifaceted nature of climate and soil datasets (Fig. 9).

3.5 Convolutional neural network (CNN)

CNN model is designed to capture spatial dependencies in time-series data through a series of specialized layers (Li et al. 2021). The model begins with an input layer, shaped to handle the temporal sequence of features. This is followed by a Convolutional Layer with 64 filters that applies convolution operations to extract local patterns from the data, which are crucial for identifying relevant features. Next, a MaxPooling Layer reduces the dimensionality of the data, focusing on the most significant features detected by the convolutional filters. The output is then flattened into a single vector, which is processed by a fully connected (Dense) layer with 50 units and a Dropout layer to prevent overfitting. Finally, the model outputs predictions through another Dense layer with a softmax activation function, classifying the data into one of six categories. The model is compiled with the Adam optimizer and trained using categorical cross-entropy loss, ensuring robust performance in classifying drought scores. The number of layers and filter sizes are selected based on empirical experiments to balance feature extraction capacity and computational efficiency. Hyperparameters are optimized by evaluating performance across different architectures, ensuring a trade-off between model complexity and generalization.

3.6 Long short-term memory (LSTM)

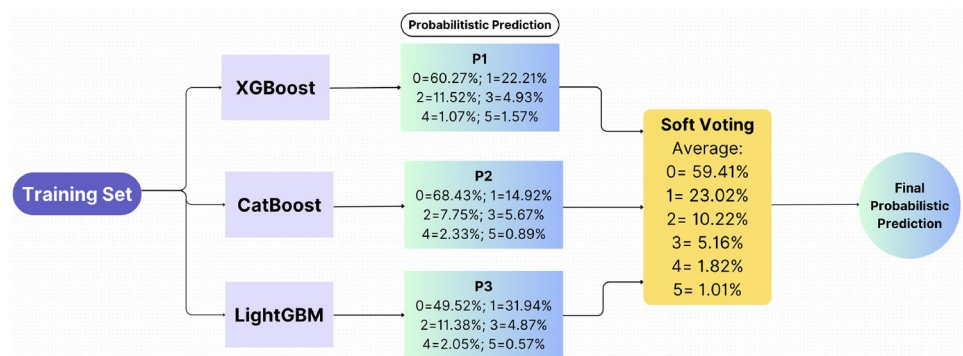
The LSTM model employed for predicting drought scores effectively captures temporal dependencies in time series data (Yu et al. 2019). Initially, the input data is preprocessed to create sequences using a sliding window approach, which is

essential for the LSTM to learn patterns over time. The model architecture comprises two LSTM layers, each with 50 units, to capture both short-term and long-term dependencies. Dropout layers with a 20 percent rate are included to prevent overfitting by randomly omitting some of the outputs of the LSTM units during training. Finally, a dense layer with a softmax activation function produces the probability distribution over the target classes. This model is trained using the Adam optimizer and categorical cross-entropy loss, and its performance is validated on a separate test set. This structure allows the model to effectively learn and predict the drought scores based on the temporal patterns in the input features. The number of LSTM layers and units are determined through empirical testing, balancing model expressiveness with computational cost. Hyperparameters such as dropout rate and sequence length are optimized based on validation performance to ensure robust generalization.

3.7 Transformer

The Transformer model is an advanced neural network architecture. Unlike traditional models that process input data sequentially, the Transformer leverages self-attention mechanisms to capture relationships between all elements of the input sequence simultaneously. This is achieved through the Multi-Head Attention layers, which allow the model to focus on different parts of the sequence at once, improving its ability to understand context and dependencies across the entire sequence (Han et al. 2021). The model also includes feed-forward neural networks for additional processing, and Layer Normalization to stabilize and accelerate training. Dropout layers are used to prevent overfitting by randomly omitting certain neurons during training. In this configuration, the Transformer model processes input features, learns important temporal dependencies through attention mechanisms, and outputs class probabilities for drought prediction. This architecture is particularly effective in capturing complex patterns in sequential data due to its ability to model long-range dependencies and parallelize computation, making it a powerful tool for drought analysis.

Fig. 9 Voting classifier model architecture



3.8 Hybrid models

Hybrid models are ML frameworks that combine different types of algorithms or network architectures to leverage the strengths of each. By integrating various methodologies, hybrid models aim to enhance predictive performance, improve generalization, and address complex tasks more effectively than single-model approaches (Krasnopolsky and Fox-Rabinovitz 2006). Hybrid models are built upon previously optimized base models for each cluster, where hyperparameters were carefully selected to maximize performance before integration into hybrid architectures.

3.8.1 CNN–LSTM hybrid model

The CNN–LSTM hybrid model combines the strengths of CNN and LSTM networks to effectively handle spatial and temporal dependencies in the data (Vo et al. 2023). The architecture begins with an input layer that feeds data into the CNN layers. The first layer is a 1D Convolutional layer (Conv1D) which applies convolutional filters to capture spatial features within the input data. This is followed by a MaxPooling layer to reduce the dimensionality and retain the most significant features. The output from the CNN layers is then passed to an LSTM layer, which is designed to

capture the temporal dependencies in the data by maintaining a memory of previous inputs. This LSTM layer outputs sequences, which are then flattened into a single vector. This vector is processed through a series of dense (fully connected) layers. A dropout layer is included to prevent overfitting by randomly setting a fraction of input units to zero during training. Finally, the model includes a dense layer with a `softmax` activation function to produce the final classification into one of the predefined categories (Fig. 10).

3.8.2 Attention-LSTM hybrid model

The Attention-LSTM model integrates LSTM layers for capturing temporal dependencies in the data with attention mechanisms that enhance focus on important time steps. Initially, the input layer accepts the data, followed by the LSTM layer, which captures the temporal dependencies within the time series data. The multi-head attention layer then improves focus on significant time steps. The feed-forward network, including the dense layer and layer normalization, processes the data and applies normalization to ensure consistent data scaling. In the final classification stage, the flatten layer converts the processed data into a single vector, which is then handled by the first dense layer. A dropout layer is included to prevent overfitting by randomly

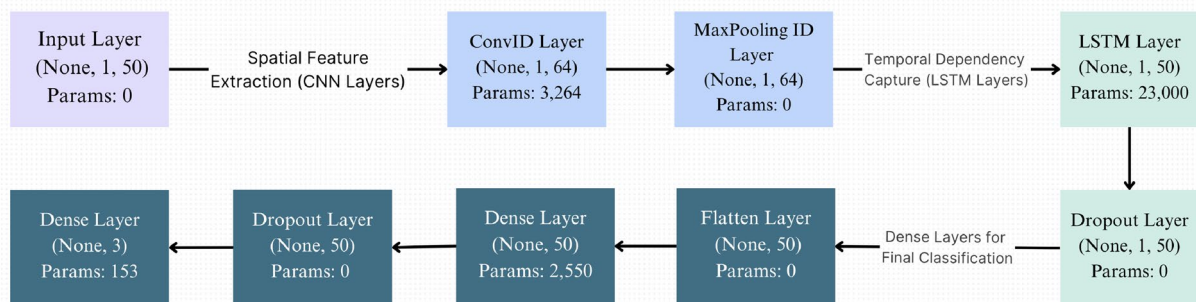


Fig. 10 CNN–LSTM hybrid model architecture

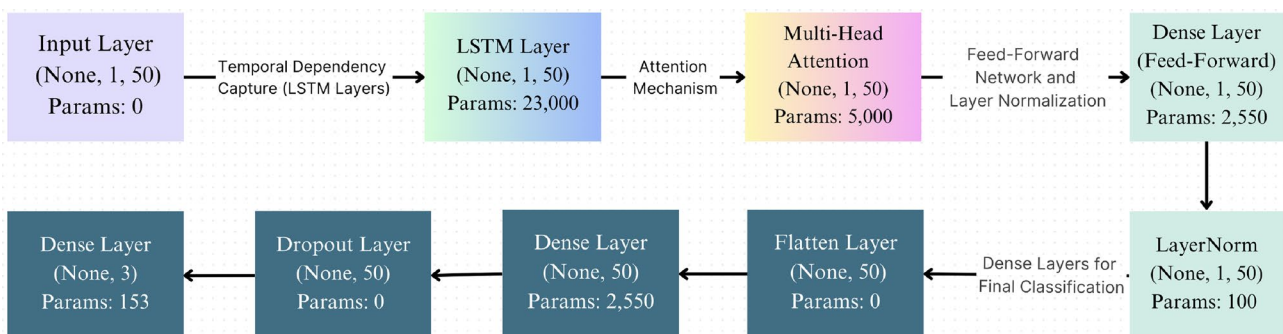


Fig. 11 Attention-LSTM hybrid model architecture

disabling some neurons. Lastly, the final dense layer produces the model's output, categorizing the data into three distinct classes (Fig. 11).

3.9 Feature importance

The feature importance plots for LightGBM, the best-performing ML model, across the three clusters highlight the key variables influencing drought scores, as shown in Figs. 12, 13 and 14. Since feature importance could not be extracted from DL models, the analysis is conducted using the best-performing ML model instead. For comparison, two separate feature importance techniques are applied: impurity-based importance and Shapley Additive Explanations (SHAP). The relationship between climate variables and drought risk across the identified clusters highlights the multifaceted influence of geographic, atmospheric, and edaphic factors on drought susceptibility. The feature importance analysis provides insights into the key factors distinguishing

each cluster, reflecting long-term climate and environmental conditions rather than short-term weather variations. Since k-means clustering is applied to categorize regions based on persistent climate and soil characteristics, the identified influential features align well with known geographical patterns. The high importance of latitude and longitude in all clusters serves as a model verification step, confirming that the clustering process effectively captures spatial dependencies inherent in climate data.

- Cluster 1, which predominantly covers the Northern and Western U.S., including parts of the Midwest and Great Plains, exhibits a strong influence of maximum temperature, specific humidity, and dew/frost point. This suggests that these regions experience significant seasonal variations, where extreme temperatures and humidity shifts play a crucial role in defining drought conditions. The presence of earth skin temperature and wet bulb temperature as key factors further indicates high exposure to heat

Fig. 12 Feature importance for cluster 1

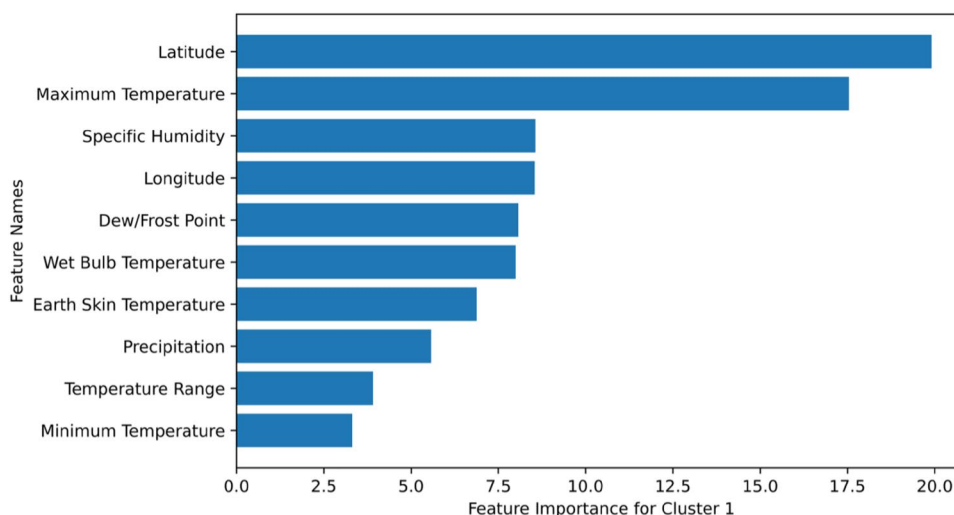


Fig. 13 Feature importance for cluster 2

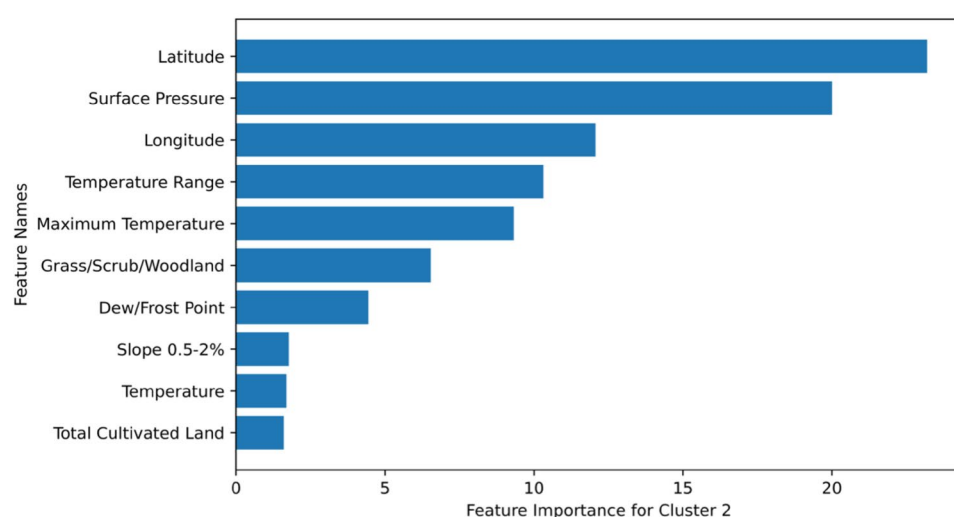
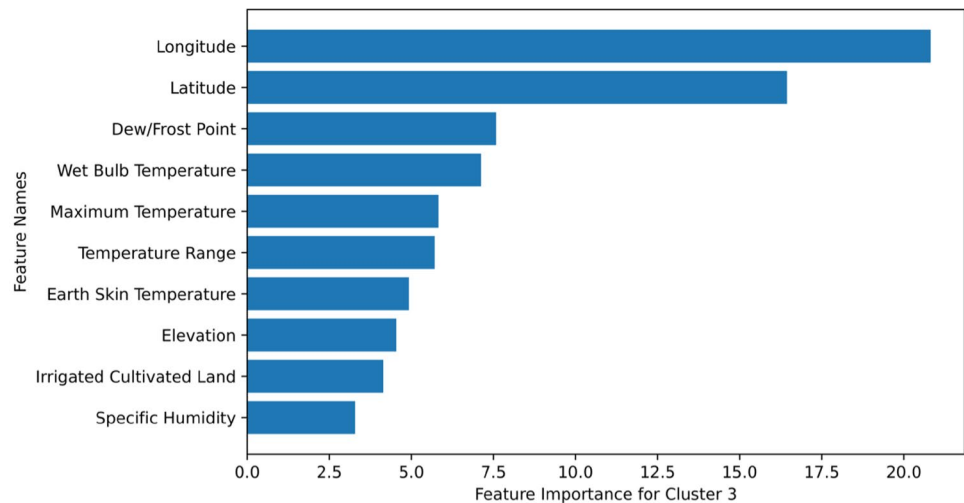


Fig. 14 Feature importance for cluster 3

stress and surface energy balance effects, which aligns with the continental climate influences characteristic of the Northern Great Plains and upper Midwest. Additionally, the moderate importance of precipitation suggests that this cluster experiences moderate to low rainfall, a trait commonly associated with semi-arid regions like the western U.S. and parts of the Midwest.

- **Cluster 2**, covering the Southeastern U.S. and parts of the Eastern Midwest, is strongly influenced by surface pressure, suggesting that humidity-driven atmospheric pressure patterns play a crucial role in this region. This aligns with the Southeast and Gulf Coast states, where such climatic patterns are prominent. Additionally, the significance of temperature range and maximum temperature highlights the presence of warm, humid conditions with notable seasonal variations, particularly in states like Georgia, Alabama, and South Carolina. The inclusion of vegetation-related features, such as grassland and cultivated land, underscores the prevalence of both agricultural and natural vegetation, which is characteristic of the Eastern Midwest and the Southeastern agricultural zones.
- **Cluster 3**, which spans the Central and Western U.S., including the Southwest and parts of the Rocky Mountains, is characterized by the strong influence of dew/frost point and wet bulb temperature. This highlights the role of dry-air dynamics and heat stress adaptation, which are critical factors in arid and semi-arid regions such as the Southwest and interior West. The significance of elevation further confirms that this cluster encompasses topographically diverse areas, including the Rocky Mountains and western highlands, where altitude plays a key role in shaping temperature and precipitation patterns. Additionally, the importance of irrigated cultivated land suggests that human-modified landscapes, particularly in water-limited areas like California's Central Valley and parts of Arizona, are integral to defining this region.

The SHAP summary plots for LightGBM model across the three clusters, shown in Fig. 15, provide a detailed insight into how various features influence drought risk across different regions. Each cluster is characterized by distinct features that highlight the complex interplay between climatic, geographic, and soil factors affecting drought conditions. The feature importance results reveal that latitude and longitude are among the most influential features across all three clusters. This consistent pattern aligns with domain expectations, as geographic location plays a critical role in regional climate behavior and drought susceptibility. The significance of these spatial variables serves as a form of model verification, suggesting that the models are capturing meaningful and interpretable relationships.

- **Cluster 1:** Key features include maximum temperature, specific humidity, earth skin temperature, and precipitation. Higher maximum temperatures and earth skin temperatures are associated with more severe drought conditions. In contrast, higher specific humidity and precipitation are linked to lower drought risk, underscoring the role of atmospheric moisture. Dew/frost point and wind speed also contribute, highlighting the importance of dynamic weather conditions.
- **Cluster 2:** Important predictors include surface pressure, maximum temperature, temperature range, and soil-related features such as grass/scrub/woodland and nutrient retention. Higher surface pressure is linked to more stable and dry conditions. Wide temperature fluctuations and degraded soil health (e.g., poor nutrient retention) are associated with elevated drought risk, while resilient vegetation cover may mitigate these effects.
- **Cluster 3:** Influential features include dew/frost point, wet bulb temperature, nutrient availability, and irrigated cultivated land. Lower dew points and higher wet bulb

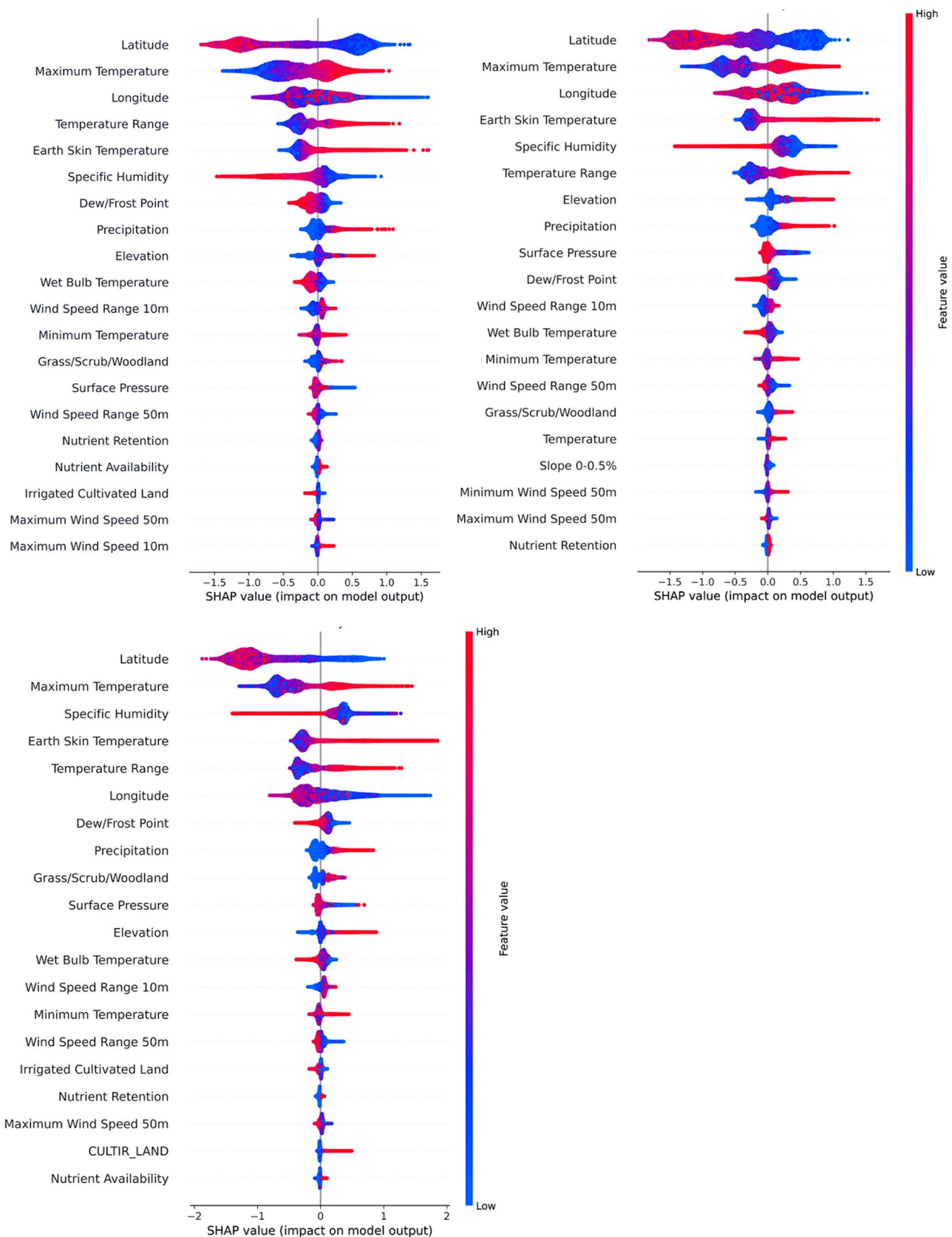


Fig. 15 SHAP summary plots for each cluster

temperatures contribute to drought severity, reflecting more stressful moisture and heat conditions for vegetation. Soil features such as nutrient availability and irrigation practices emerge as critical factors influencing drought vulnerability in these regions.

Across all clusters, features such as temperature range, elevation, and various soil properties consistently appear, highlighting their universal importance in drought prediction. For example, higher temperature ranges can stress plants and reduce water availability, while higher elevations might experience different precipitation patterns that affect drought conditions. Soil properties such as nutrient retention

and availability are vital; soils with better nutrient profiles support more resilient plant growth, potentially reducing drought severity. Conversely, poor soil conditions can exacerbate drought impacts by limiting vegetation growth and water retention.

3.10 Comparison of algorithms

Table 6 illustrates the performance metrics-Accuracy, Recall, Precision, and F1 Score-of various ML and DL models across three clusters for drought prediction.

The DL models generally exhibit superior performance compared to traditional ML models due to their ability to

Table 6 Test set performance metrics for algorithms across clusters

	Accuracy	Recall	Precision	F1 Score	Comp. time (s)
<i>Cluster 1</i>					
XGBoost	0.6508	0.6508	0.5566	0.5778	120
CatBoost	0.6645	0.6823	0.5724	0.5499	140
LightGBM	0.6573	0.6573	0.5618	0.5838	110
Stacking	0.7615	0.7635	0.7615	0.6362	200
Voting classifier	0.7927	0.7813	0.7925	0.6942	180
CNN	0.8295	0.8021	0.8288	0.7954	600
LSTM	0.9515	0.9180	0.9515	0.9344	900
Transformer	0.9224	0.9118	0.9219	0.9133	1200
CNN-LSTM hybrid model	0.8942	0.8528	0.8698	0.8650	1000
Attention-LSTM hybrid model	0.9623	0.9371	0.9642	0.9530	1300
<i>Cluster 2</i>					
XGBoost	0.5938	0.5939	0.6285	0.6059	115
CatBoost	0.6824	0.6912	0.5326	0.5726	135
LightGBM	0.6800	0.6786	0.5622	0.5997	108
Stacking	0.7808	0.7808	0.6563	0.6689	195
Voting classifier	0.7798	0.7721	0.6131	0.6618	175
CNN	0.7946	0.7798	0.7913	0.7261	580
LSTM	0.9204	0.8856	0.9204	0.8838	890
Transformer	0.9783	0.9511	0.9736	0.9647	1180
CNN-LSTM hybrid model	0.8651	0.8319	0.8649	0.8441	980
Attention-LSTM hybrid model	0.9623	0.9371	0.9642	0.9530	1250
<i>Cluster 3</i>					
XGBoost	0.6860	0.6860	0.5797	0.6165	118
CatBoost	0.7072	0.7071	0.5766	0.5942	138
LightGBM	0.7078	0.7069	0.5942	0.6263	112
Stacking	0.8041	0.7466	0.8041	0.6993	198
Voting classifier	0.8215	0.7919	0.8295	0.7176	178
CNN	0.8191	0.8060	0.8187	0.7664	590
LSTM	0.9318	0.8777	0.9318	0.9017	910
Transformer	0.9386	0.9099	0.9289	0.9243	1190
CNN-LSTM hybrid model	0.8664	0.8521	0.8647	0.8511	970
Attention-LSTM hybrid model	0.9702	0.9497	0.9698	0.9614	1270

The bold values indicate the highest performance metric (Accuracy, Recall, Precision, and F1 Score) achieved by any model within each cluster. For each metric, the best-performing model in a given cluster is highlighted in bold to emphasize its relative superiority

capture both spatial and temporal dependencies in the data. Given that our dataset has a spatio-temporal structure, models that effectively handle both dimensions yield the best results. Specifically, recurrent architectures such as LSTM are highly effective in capturing long-term sequential dependencies, making them particularly suitable for drought prediction, as climatic conditions evolve gradually over time. On the other hand, attention-based models, such as Transformers, excel at capturing spatial dependencies by dynamically focusing on the most relevant features across different regions. The hybrid Attention-LSTM model combines the strengths of both approaches, allowing it to model both temporal patterns and spatial variations in the data simultaneously. As a result, this model effectively learns how drought conditions progress over time while also incorporating the influence of geographical characteristics. As demonstrated in Table 6, the Attention-LSTM model consistently outperforms all other approaches, highlighting its capability to model the complex interactions between spatial and temporal factors in drought prediction.

- Cluster 1: The Attention-LSTM hybrid model achieves the highest performance with an accuracy of 0.9623 and an F1 Score of 0.9530, demonstrating strong capability in modeling long-term dependencies and dynamically weighing feature importance. The LSTM model offers slightly lower but comparable results, with the added advantage of shorter computational time, making it a practical alternative when time efficiency is critical. Among traditional ML models, the Voting Classifier performs best, with an accuracy of 0.7927 and an F1 Score of 0.6942, though it lags behind DL models due to its limited ability to capture sequential dependencies.
- Cluster 2: The Transformer model achieves the best performance, with an accuracy of 0.9783 and an F1 Score of 0.9647, leveraging self-attention mechanisms to effectively model long-range dependencies in climate patterns.
- Cluster 3: The Attention-LSTM hybrid model again leads in performance with an accuracy of 0.9702 and an F1 Score of 0.9614. The Voting Classifier stands out among traditional ML methods in this cluster, achieving an accuracy of 0.8215 and an F1 Score of 0.7176, but falls short in handling the complexity of temporal drought dynamics.

However, while DL models yield superior predictive performance, they come at the cost of significantly higher computation time. The DL models are trained with 100 epochs, resulting in notable improvements in accuracy and recall, but require considerably longer computational time compared to traditional ML models. For instance, the Attention-LSTM hybrid model requires 1300s to train, whereas XGBoost only takes 120 s, despite its lower performance. This computational trade-off is crucial for practical applications where

model efficiency and inference speed are critical factors. A comparison of computation time versus performance is summarized in Table 6, highlighting the efficiency-performance trade-off between different models.

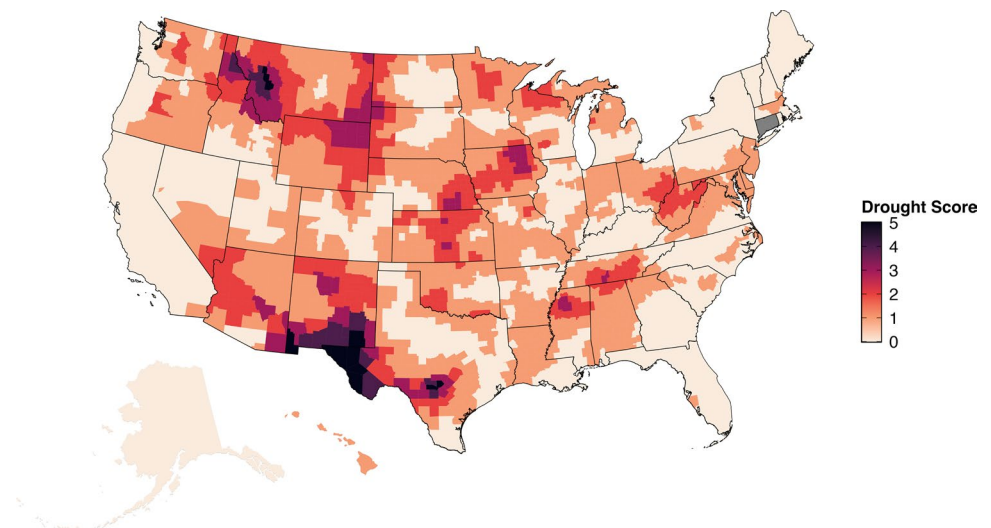
The metrics presented are weighted averages to address class imbalance in the drought scores (Winkler and Clemen 1992) (*Please refer to Appendix 3 for detailed formulas*). Oversampling techniques are applied to mitigate this imbalance, but the improvements in model performance are not substantial. This highlights that while DL models excel in accuracy and predictive capability, their computational demands and sensitivity to class imbalances require careful consideration.

A key advantage of our approach is its adaptability to new data. As additional climate and soil data become available, our model can be seamlessly updated and retrained, ensuring its predictive accuracy remains robust under changing conditions. This flexibility is crucial for real-world applications, as climate patterns shift over time due to factors such as global warming and land-use changes. By continuously integrating new observations, the model can provide more reliable and up-to-date drought predictions, aiding decision-makers in water resource management, agriculture, and disaster preparedness. Furthermore, this framework can be extended to different regions by fine-tuning the model with localized data, making it a scalable and versatile solution for drought forecasting in various climatic conditions.

3.11 Drought prediction

After evaluating and selecting the most effective model for each individual region, the top-performing models from these three regions are aggregated to predict the average drought severity across the U.S. for the year 2024. The predicted drought levels for 2024 are illustrated in Fig. 16. The 2024 U.S. Drought Severity Map reveals significant regional variations in drought conditions across the country. The Southwest, particularly Texas and New Mexico, faces severe to extreme drought levels, indicated by dark purple hues, while Arizona and parts of Oklahoma experience moderate to severe drought conditions. In contrast, the West Coast states such as California, Oregon, and Washington primarily exhibit low drought severity, represented by light beige to orange hues. The Midwest and Great Plains, including states such as Kansas, Nebraska, and Iowa, display patches of moderate drought. The Northeast and Mid-Atlantic regions show predominantly low drought levels, while parts of the Southeast, particularly Mississippi and Georgia, experience moderate drought severity in localized areas. This map highlights the diverse impact of drought across different regions, influenced by geographic and climatic factors, underscoring

Fig. 16 Predicted drought levels for 2024



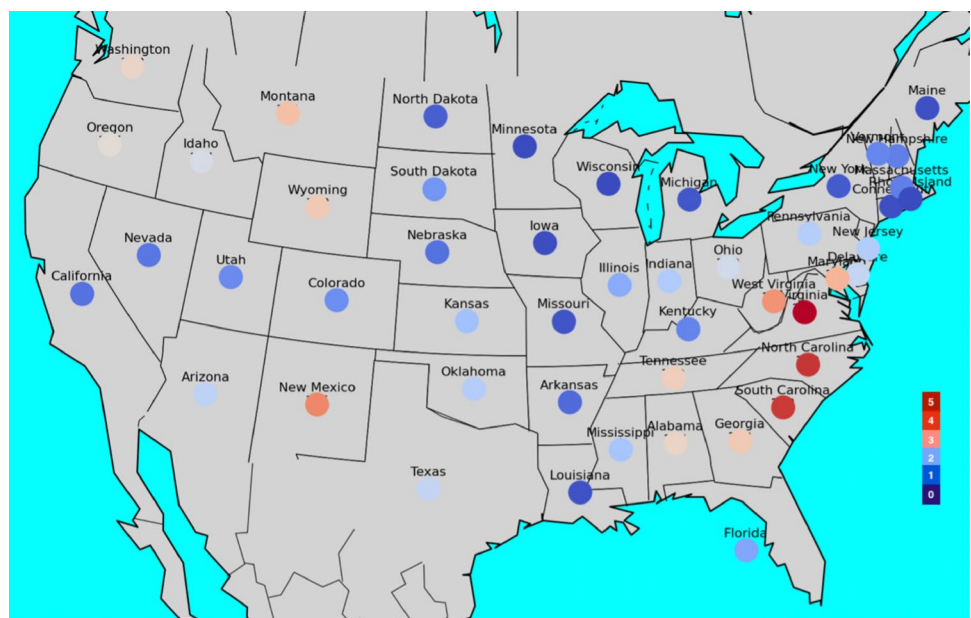
the critical need for tailored water management and climate adaptation strategies.

Using Monte Carlo Dropout, we computed the 95% prediction intervals to quantify uncertainty in drought score predictions across different regions. The results indicate varying levels of uncertainty across clusters: Cluster 1 (Western Regions), characterized by an arid and highly variable climate, has the widest prediction interval (2.10, 4.90), reflecting high uncertainty in drought severity. Cluster 2 (Eastern & Southeastern Regions), with a more stable and humid climate, shows the narrowest interval (0.95, 2.70), indicating higher model confidence in predictions. Cluster 3 (Central & Northern Plains), which includes key agricultural areas, has a moderate interval (1.50, 3.90), suggesting a balanced level

of uncertainty due to seasonal fluctuations. These results highlight that drought predictions are most uncertain in arid western regions, while the model performs more reliably in humid eastern regions.

Figure 17 illustrates the distribution of drought conditions across the U.S. during June and July 2024. The map reveals notable regional variation in drought severity. Severe drought is evident in states like South Carolina and Georgia, which have Drought Scores of 5 and 4, marked by deep red shades-indicating extremely dry conditions. States such as Colorado, Kansas, and Florida show moderate drought levels with scores ranging from 2 to 3, represented by yellow to light blue markers. On the other hand, California, New York, and Texas report milder drought with scores of

Fig. 17 Predicted drought levels for the recent months of June and July in 2024



1, suggesting limited dryness. Finally, areas like Oregon and Idaho display minimal or no drought, with Drought Scores of 0, highlighted in very light tones. Overall, while the Southeast is facing severe dryness, other regions show moderate to minimal impact, providing a snapshot of early summer drought patterns in 2024.

4 Conclusion and discussion

This study provides a comprehensive analysis of drought conditions in the U.S., emphasizing the intricate relationships between climatic variables, soil properties, land use, and meteorological factors. By integrating ML and DL models with a clustering approach, we identified key environmental drivers of drought across different regions. The findings underscore the importance of integrating diverse datasets and employing advanced modeling techniques to develop accurate and robust drought prediction systems.

The comprehensive analysis of drought conditions and their correlation with environmental factors across various regions in the U.S. underscores the intricate interplay between climate, soil properties, land use, and weather patterns in influencing drought susceptibility.

- Areas with higher drought scores, particularly in states such as Nevada, Arizona, and parts of New Mexico, exhibit specific characteristics such as lower humidity, decreased precipitation, elevated temperatures, and increased wind speeds, which collectively exacerbate drought conditions.
- The relationship between land use patterns and drought vulnerability reveals that regions with less cultivated land and more grassland are more prone to drought, suggesting that certain types of vegetation and land management practices may mitigate or intensify drought impact.
- The findings also highlight the role of soil properties, where soils in drought-prone areas tend to have better workability and rooting conditions but may lack in water retention and nutrient availability compared to non-drought-affected soils.
- Clustering the regions into three distinct categories based on climatic and soil characteristics helped to identify unique environmental factors influencing drought susceptibility in each cluster.
 - *Cluster 1* Primarily influenced by maximum temperature and specific humidity, highlighting the vulnerability of **western and northern** regions with significant seasonal temperature variations and low humidity levels. Higher temperatures exacerbate drought severity, while higher specific humidity can mitigate it. Additional factors such as earth skin temperature and precipitation also play significant roles in this cluster, underscoring the critical impact of atmospheric conditions.
 - *Cluster 2* Characterized by surface pressure and longitude, encompassing **central U.S. and some interior western regions** with continental climates and variable precipitation patterns. The importance of surface pressure indicates that stable atmospheric conditions can lead to prolonged droughts. Soil-related features such as grass/scrub/woodland and nutrient retention are also crucial, suggesting that land cover and soil health significantly impact drought vulnerability. Regions with better vegetation cover and nutrient-rich soils are more resilient to drought.
 - *Cluster 3* Influenced by dew/frost point, covering eastern and southeastern regions where high temperatures and uneven rainfall distribution contribute to drought risk. Dew/frost point and wet bulb temperature are essential indicators of atmospheric moisture content, which directly affects soil moisture retention and evapotranspiration. In Cluster 3, where soil-related factors such as irrigated cultivated land and specific humidity are important, optimizing irrigation scheduling based on soil moisture monitoring and adopting precision agriculture techniques can significantly improve water use efficiency (Qadir et al. 2021). Additionally, contour farming and terracing in elevated areas can help reduce soil erosion and improve water infiltration (Tran et al. 2017).
- The analysis of feature importance based on impurity-based methods for the LightGBM model, the best performing ML model, across the clusters further reinforces these findings. In Cluster 1, climatic factors dominate, while in Clusters 2 and 3, soil properties and land use patterns are more significant. The inclusion of variables such as nutrient availability, grassland cover, and elevation in the SHAP plots emphasizes the need for a comprehensive approach that considers both climatic and edaphic factors in drought prediction and management.
- The Granger causality analysis confirms that wind speed, temperature, and precipitation strongly predict drought severity:
 - Maximum wind speed at 50 m ($p < 10^{-300}$) and at 10 m ($p < 10^{-295}$) are significantly linked to increasing drought severity.
 - Air temperature at 2 m ($p < 10^{-300}$), maximum air temperature at 2 m ($p < 10^{-300}$), and minimum air temperature at 2 m ($p < 10^{-300}$) show strong causal relationships with drought progression.

- Precipitation ($p < 10^{-141}$) and specific humidity ($p < 10^{-63}$) are major determinants of drought occurrence.
- The performance metrics of various ML and DL models across the clusters demonstrate the superior predictive capabilities of DL models, particularly the Attention-LSTM hybrid model and Transformer model. These models exhibit high accuracy and F1 scores, with strong alignment between model predictions and the observed drought severity patterns in the 2024 forecast.
- Extending the study beyond the U.S. to regions with diverse climate regimes, ensuring the generalizability and adaptability of the proposed methodology across various geographical and environmental contexts.
- Investigating the integration of physics-informed ML techniques to embed domain knowledge into AI-driven drought forecasting, improving model interpretability and reliability.
- Enhancing the scalability and efficiency of models to enable real-time operational drought forecasting, thereby facilitating their deployment in early warning systems and proactive water resource management strategies.

While this study provides valuable insights into drought prediction and its environmental drivers, certain limitations should be acknowledged:

- The dataset utilized spans from 2000 to 2020, with more recent data (2021–2024) excluded from model validation due to data availability constraints. Incorporating up-to-date datasets in future studies will enhance the robustness and accuracy of model validation.
- The high computational cost associated with DL models remains a challenge. Future research should focus on optimizing these models for efficiency without compromising predictive performance.

To further advance the field of AI-driven drought prediction, future research should explore the following directions:

- Developing hybrid models that integrate socio-economic factors-such as water consumption trends, irrigation practices, and land use changes-alongside meteorological and soil parameters to enhance drought prediction accuracy and decision-making frameworks.

This study demonstrates the effectiveness of AI-driven drought prediction and provides a foundation for further advancements in regional drought forecasting. The methodology developed here can be adapted to similar regions

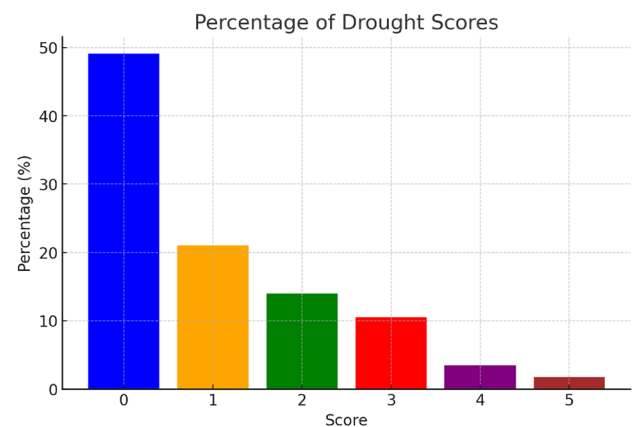


Fig. 18 Percentage of drought scores

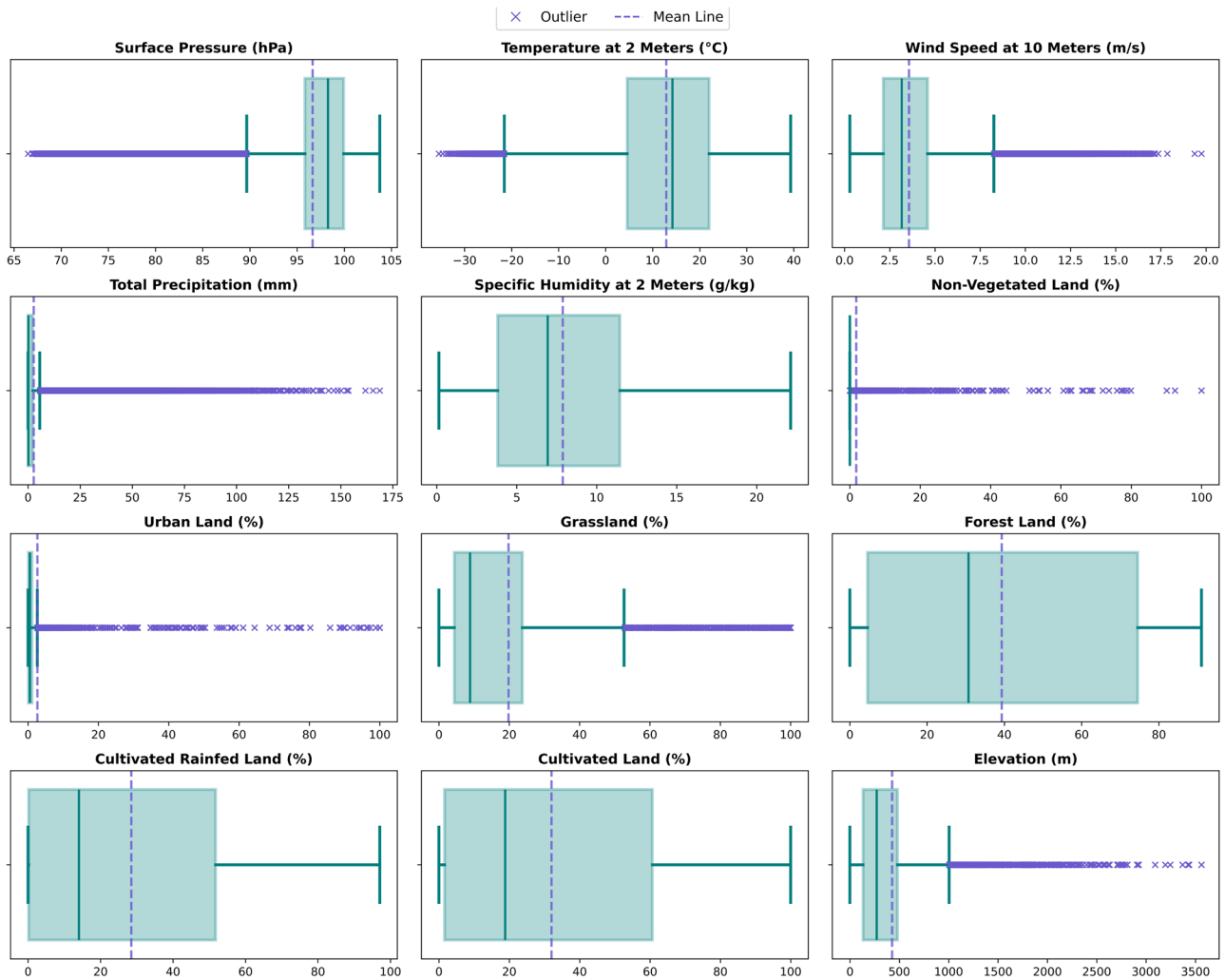


Fig. 19 Boxplot of climate and soil features

worldwide, offering valuable insights for policymakers and environmental agencies in designing sustainable drought mitigation strategies.

Appendix 1

Percentage of drought scores for U.S. in the dataset is provided below (Fig. 18).

Appendix 2

See Fig. 19.

The boxplot visualization displays the distribution of various environmental features, highlighting the presence of outliers (marked as “x”) in variables. The spread of data across different features shows significant variability,

particularly in precipitation and elevation, where extreme values are more prevalent.

Appendix 3

To classify a 0 to 5 drought score using weighted averages for accuracy, sensitivity, specificity, and F1 score, the following formulas with definitions for true positives (TP), false negatives (FN), false positives (FP), and true negatives (TN) are used.

Accuracy:

$$\text{Accuracy} = \frac{\sum_{i=0}^5 TP_i}{\sum_{i=0}^5 (TP_i + FP_i + FN_i)}$$

Sensitivity (Recall) for each class:

$$\text{Sensitivity}_i = \frac{TP_i}{TP_i + FN_i}$$

Specificity for each class:

$$\text{Specificity}_i = \frac{TN_i}{TN_i + FP_i}$$

F1 Score for each class:

$$\text{F1 Score}_i = \frac{2 \cdot TP_i}{2 \cdot TP_i + FP_i + FN_i}$$

Weighted Average Accuracy:

$$\text{Weighted Average Accuracy} = \sum_{i=0}^5 \left(\frac{N_i}{N} \cdot \text{Accuracy}_i \right)$$

Weighted Average Sensitivity (Recall):

$$\text{Weighted Average Sensitivity} = \sum_{i=0}^5 \left(\frac{N_i}{N} \cdot \text{Sensitivity}_i \right)$$

Weighted Average Specificity:

$$\text{Weighted Average Specificity} = \sum_{i=0}^5 \left(\frac{N_i}{N} \cdot \text{Specificity}_i \right)$$

Weighted Average F1 Score:

$$\text{Weighted Average F1 Score} = \sum_{i=0}^5 \left(\frac{N_i}{N} \cdot \text{F1 Score}_i \right)$$

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Data availability The dataset is openly accessible to the public and can be found at U.S. Drought Monitor [https://droughtmonitor.unl.edu/Data.aspx].

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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